

Hypothesis Testing:

This is our third major unit of inference.

Recall:

- Point Estimation — Aims to give one sensible value for a population parameter.
- Confidence Sets — Aims to give a range of likely to contain a population parameter.

In **Hypothesis Testing**, our goal is to answer Yes/No, as to whether a particular statement is supported by the data.

→ We call such statements **statistical hypotheses**.

Def. (Statistical Hypotheses): A claim about:

- i) the value(s) of an unknown population parameter
- ii) the form of a population distribution. || We will not see this kind of hypotheses in this class.

Examples of Hypotheses:

- i) [• "The probability of this coin landing heads-up is $1/2$ "
- ii) [• "The returns of the S&P are normally distributed"

We will test these hypotheses by seeing how strongly our observed data supports (or contradicts) them.

Motivation: Coin Flip

Let $X_1, \dots, X_{10} \stackrel{iid}{\sim} \text{Ber}(p)$, with unknown p .

- $X_i = 1$ means the i^{th} flip was a Heads

Hypothesis: $p = 1/2$.

Consider whether you (intuitively) believe our hypothesis is true if we actually observe:

- ★ • All 10 flips are "H" $\rightarrow$ Very unlikely outcome if $p = 1/2$ (has prob. 0.1%)
- 6 flips are "H" $\rightarrow$ Somewhat likely outcome if $p = 1/2$ (prob. 21%)
- 5 flips are "H" $\rightarrow$ Somewhat likely outcome if $p = 1/2$ (prob. 25%)

But now imagine we flipped this coin 100 times:

- ★ • All 100 are "H" $\rightarrow$ If $p = 1/2$, occurs w/ prob. ≈ 0
 - ★ • 60 "H"s $\rightarrow$ If $p = 1/2$, occurs w/ prob. $\approx 2 \times 10^{-4}$
 - 50 "H"s $\rightarrow$ If $p = 1/2$, occurs w/ prob. $\approx 8\%$
- || Come from the p.m.f. of a Binomial $(100, 1/2)$ r.v.

★: These outcomes really seem to contradict our hypothesis that $p = 1/2$, since we're very unlikely to see these outcomes "by chance" if p did equal $1/2$.

$\rightarrow$ Hence we might reject the hypothesis that $p = 1/2$

Takeaways:

1) We think of rejecting a hypothesis if our observations seem incompatible (i.e. very unlikely) with the hypothesis.

2) Whether we reject a hypothesis depends on the sample size and the realizations we obtained.

$\rightarrow$ Before we obtain realizations, to reject (or not) is random.

[Think of a point estimate $\hat{\theta}(x_1, \dots, x_n)$ vs. an estimator $\hat{\theta}(X_1, \dots, X_n)$]

We'll now try to formalize these two ideas.

Framework:

Def. (Null Hypothesis, H_0): A statistical hypothesis that we initially assume is true, i.e. the "status quo". E.g.'s that a coin is 'fair', or that some new pharmaceutical drug has no effect. Think: "starting out skeptical"

Def. (Alternative Hypothesis, H_A): Another statistical hypothesis that is mutually exclusive with the Null H_0 . E.g.'s that a coin is not fair, or this drug has some effect.

These two hypotheses come in pairs, and (for our purposes) need not be collectively exhaustive.

Examples:

- $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Ber}(p)$. $H_0: p = 1/2$, $H_A: p \neq 1/2$
- $Y_1, \dots, Y_n \stackrel{iid}{\sim} N(\mu, 2)$. $H_0: \mu = 0$, $H_A: \mu > 0$

These must be specified by us, the researcher, or given in a particular problem.

Goal: Let $X_1, \dots, X_n$ be random variables with unknown pop parameter θ .

Given null hypothesis H_0 , and alternative hypothesis H_A , build a test

$\mathcal{H}(X_1, \dots, X_n)$ that equals:

- 0 if we cannot reject H_0 || "The data doesn't strongly contradict H_0 "
- 1 if we can reject H_0 || "The data does strongly contradict H_0 "

By construction $\mathcal{H}(X_1, \dots, X_n)$ will be random, and we aim to define it so it has "nice" properties.

Types of Errors:

Recall - the parameter θ is assumed to be fixed.

→ So, H_0 is actually true or false.

But $H(X_1, \dots, X_n)$ may equal either 1 or 0, at random.

→ So, we can choose to reject H_0 or not to.

What are the possible outcomes?

		<u>Ground Truth</u>	
		H_0 true	H_0 false
<u>Test Outcome</u>	Not Reject H_0 ($H(X_1, \dots, X_n) = 0$)	✓	✗ (Type II Error)
	Reject H_0 ($H(X_1, \dots, X_n) = 1$)	✗ (Type I Error)	✓

↳ These probabilities must add to 1!

Two types of errors we might make:

- Type I Error: Reject H_0 , when it is actually true - a false positive
- Type II Error: Do not reject H_0 , when it is actually false - a false negative

Note: There is a tradeoff between these two types of error.

- If $H(X_1, \dots, X_n) = 0$ always, we never make a Type I error, but will always make a Type II error when H_0 is false.
- If $H(X_1, \dots, X_n) = 1$ always, vice-versa. Always make a Type I error when H_0 is true.

Since $H(X_1, \dots, X_n)$ is random (outside of these extreme examples), we care about the probability of making Type I / Type II errors.

Def. (Size/Significance Level, Power): Given H_0, H_1 , and a test $H(X_1, \dots, X_n)$:

- the size of our test is: $IP(\text{Type I Error}) = IP(H(X_1, \dots, X_n) = 1 \mid H_0 \text{ is true})$
(or significance level)

- the power of our test is: $1 - IP(\text{Type II Error}) = 1 - IP(H(X_1, \dots, X_n) = 0 \mid H_0 \text{ is false})$

We want the size of a test to be small, and its power to be large. If a test of size α rejects H_0 , we say " H_0 is rejected at α level".

In practice, we often specify the size of a desired test, e.g. 5%. We want this to be small so we avoid making "false discoveries" - hence why H_0 represents the "status quo".

Think: "innocent until proven guilty"

Test Statistics + Rejection Region:

We will construct a test $H(X_1, \dots, X_n)$ by combining both:

- i) a test statistic, $T(X_1, \dots, X_n)$, e.g. $\bar{X}_n$

- ii) a rejection region, R , e.g. $[1, \infty)$, or $(-\infty, -1] \cup [1, \infty)$

such that:

$$H(X_1, \dots, X_n) = \begin{cases} 1 & \text{if } T(X_1, \dots, X_n) \in R \\ 0 & \text{otherwise.} \end{cases}$$

This is very useful to do! Because it lets us express the size of a test as:

$$\begin{aligned} \text{Size}(H) &= IP(H(X_1, \dots, X_n) = 1 \mid H_0 \text{ is true}) \\ &= \underline{IP(T(X_1, \dots, X_n) \in R \mid H_0 \text{ is true})} \end{aligned}$$

→ If we know the distribution of our test statistic $T(X_1, \dots, X_n)$ when H_0 is true, we can compute this probability for all sorts of rejection regions!

Building a Test Statistic:

Most frequently (in this class), the null and alternative hypotheses will be about the mean in a population.

I.e., $H_0: \mu = \mu_0$, where μ is the unknown mean of $X_1, \dots, X_n$, and μ_0 is some given value.

Consider: If this H_0 is true, then the pop. mean is μ_0 .

→ Hence, by CLT (or its variants):

$$\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} \sim N(0, 1), \text{ when } n > 30$$

$$\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S(X_1, \dots, X_n)} \sim N(0, 1), \text{ when } n > 40, \text{ and } S^2(X_1, \dots, X_n) = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

Only holds
under H_0
(i.e. when
 H_0 is true)

So, we let our test statistic be equal to the number of standard deviations (of $\bar{X}_n$) between the sample mean $\bar{X}_n$ and the pop. mean under H_0 .

$$\sim T(X_1, \dots, X_n) = \begin{cases} \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma} & \text{if } \sigma \text{ is known} \\ \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{S} & \text{otherwise} \end{cases}$$

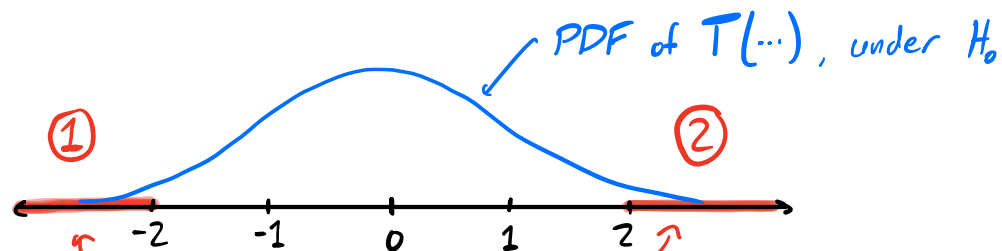
Putting this all together: "Under H_0 , $T(X_1, \dots, X_n) \overset{\text{approx}}{\sim} N(0, 1)$ for large n "

Building a Rejection Region:

Recall that $T(X_1, \dots, X_n) \in R$ corresponds to rejecting the null hypothesis H_0 .

So we now want to think: what values of the test statistic $T(\dots)$ will strongly contradict H_0 in favor of H_A ?

We know that under H_0 , $T(\dots) \sim N(0,1)$.



If H_0 is true, values in these regions are very unlikely

①: Values here contradict $H_0: \mu = \mu_0$, in favor of:

- $H_A: \mu \neq \mu_0$
- $H_A: \mu < \mu_0$ (Small $T \Leftrightarrow \bar{X}_n < \mu$)

But they would not be in favor of $H_A: \mu > \mu_0$.

②: Values here also contradict $H_0: \mu = \mu_0$, but in favor of:

- $H_A: \mu \neq \mu_0$
- $H_A: \mu > \mu_0$ (Large $T \Leftrightarrow \bar{X}_n > \mu_0$)

not $H_A: \mu < \mu_0$

$\Rightarrow$ The rejection region should be built from the tails of our test statistic $T(\dots)$'s distribution, and the specifics of our alternative hypothesis H_A .

H_A	Rejection Region
$\mu \neq \mu_0$	$(-\infty, -c] \cup [c, \infty)$
$\mu > \mu_0$	$[c, \infty)$
$\mu < \mu_0$	$(-\infty, -c]$

for some (yet to be determined) constant $c > 0$.

Recall: $\text{Size}(\mathcal{H}) = P(T(X_1, \dots, X_n) \in R \mid H_0 \text{ is true})$

To build a test $\mathcal{H}(X_1, \dots, X_n)$ of size α , for some given α , we must choose the constant c above (and hence a rejection region R) s.t.

$$P(T(X_1, \dots, X_n) \in R \mid H_0 \text{ is true}) = \alpha$$

$H_A: \mu > \mu_0$

$$P(T(X_1, \dots, X_n) \in R \mid H_0 \text{ is true}) = \alpha$$

↓ by defn of R when $H_A: \mu > \mu_0$

$$P(\underline{T(X_1, \dots, X_n)} \geq c \mid H_0 \text{ is true}) = \alpha$$

↑ Has $N(0,1)$ dist'n when H_0 is true

$\Rightarrow c = z_\alpha$, the α -critical value of a std. normal!

Similarly:

• $H_A: \mu < \mu_0$, $R = (-\infty, -z_\alpha]$ for a test of size α .

• $H_A: \mu \neq \mu_0$, $R = (-\infty, -z_{\alpha/2}] \cup [z_{\alpha/2}, \infty)$ for a test of size α .

In Summary:

- $X_1, \dots, X_n$ iid with unknown mean μ and variance σ^2 , with $n > 40$
- $H_0: \mu = \mu_0$, for some given value μ_0 ($n > 30$ if σ^2 is known)
- $T(X_1, \dots, X_n) = \frac{\sqrt{n}(\bar{X}_n - \mu_0)}{s(X_1, \dots, X_n)}$ (or $\frac{\sqrt{n}(\bar{X}_n - \mu_0)}{\sigma}$ if σ^2 is known)
- For a test of size α , reject H_0 when $T(X_1, \dots, X_n) \in R$, where:

$$R = \begin{cases} [z_\alpha, \infty) & \text{if } H_A: \mu > \mu_0 \\ (-\infty, -z_\alpha] & \text{if } H_A: \mu < \mu_0 \\ (-\infty, -z_{\alpha/2}] \cup [z_{\alpha/2}, \infty) & \text{if } H_A: \mu \neq \mu_0 \end{cases}$$

Example: Coin Flip

Let $X_1, \dots, X_{100} \stackrel{iid}{\sim} \text{Ber}(p)$ random variables, with p unknown

a) $H_0: p = 1/2$, $H_A: p > 1/2$. Find a test of size 5%.

→ Since we don't know the variance of $X_1, \dots, X_{100}$, our test statistic is:

$$T(X_1, \dots, X_{100}) = \frac{\sqrt{100}(\bar{X}_{100} - 1/2)}{s(X_1, \dots, X_{100})}$$

which is approximately $N(0, 1)$ when H_0 is true (s.t. $p = 1/2$).

With $H_A: p > 1/2$, the rejection region for a test of size 5%

is:

$$R = [\overset{= 5\%}{1.65}, \infty)$$

Hence, a test of size 5% is given by:

$$\underline{H}(X_1, \dots, X_{100}) = \begin{cases} 1 & \text{if } \frac{10(\bar{X}_{100} - 1/2)}{s(X_1, \dots, X_{100})} \geq 1.65 \\ 0 & \text{otherwise} \end{cases}$$

} Equivalent to $\bar{X}_{100} - \frac{5}{10} \cdot 1.65 \geq 1/2$
Look familiar at all?

b) If we observe $x_1, \dots, x_{100}$ and $\bar{x}_{100} = 0.6$, with $s^2(x_1, \dots, x_{100}) = 0.25$, then can we reject H_0 ?

→ Compute $T(x_1, \dots, x_{100})$ and see if it lands in the rejection region!

$$T(x_1, \dots, x_{100}) = \frac{10(0.6 - 1/2)}{\sqrt{0.25}} = \frac{1}{0.5} = \underline{2} > 1.65$$

∴ Yes, we reject the null hypothesis that $p = 1/2$, at 5% level.